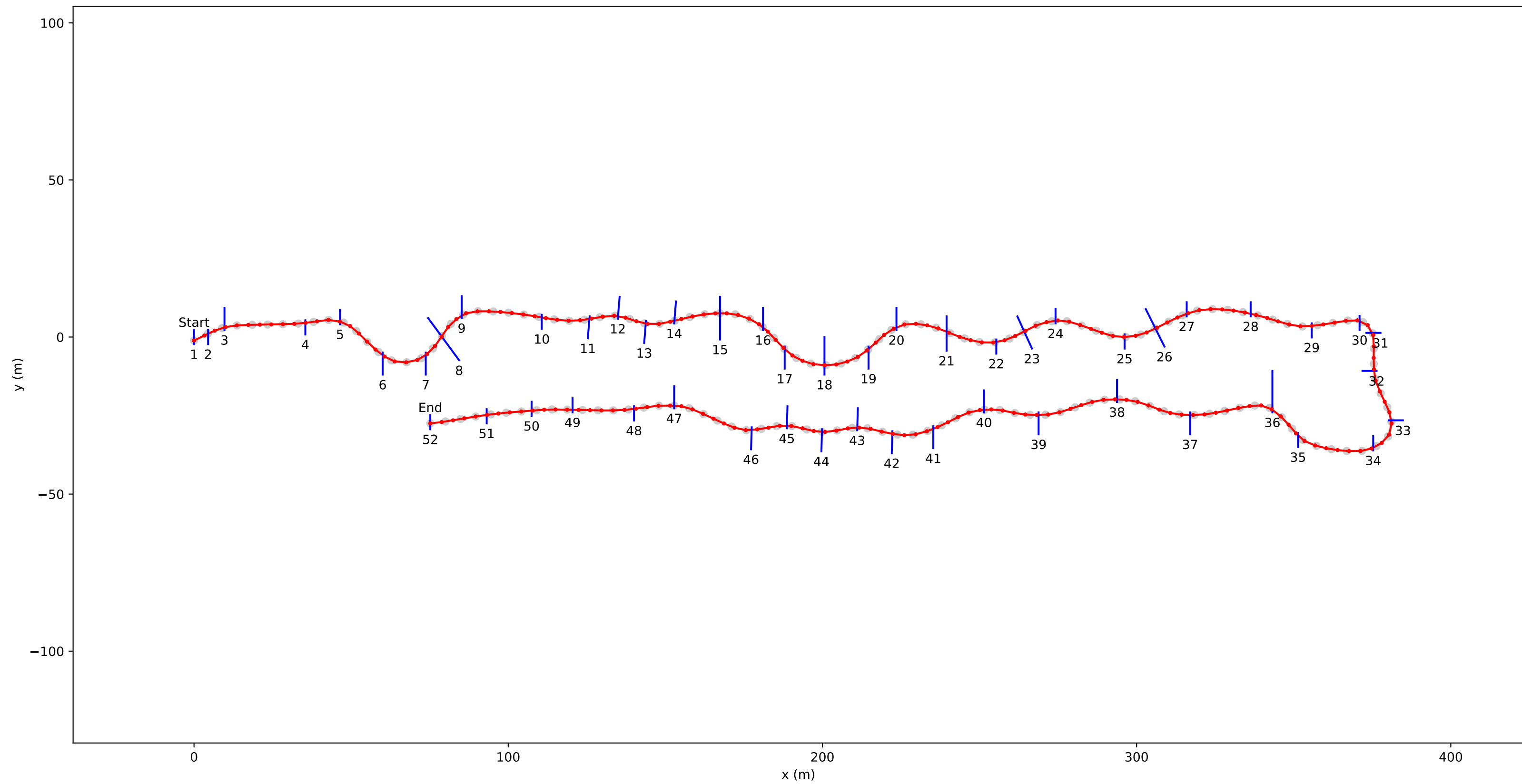


Map plot: optimized path through gates



Track Generation Report

=====
Generated: 2026-05-31 22:34:24
Gate file: ref_gates_autoX.csv
Gate count: 52
Saved DOFs file: opt_dofs_2026-05-31_22-34-23.npy

Optimization parameters

method: L-BFGS-B
maxiter: 500
ftol: 1e-07

Point counts

objective function points: 208
path generation points: 156

Results

total optimization time: 5079.172 s
total path length: 755.9129 m

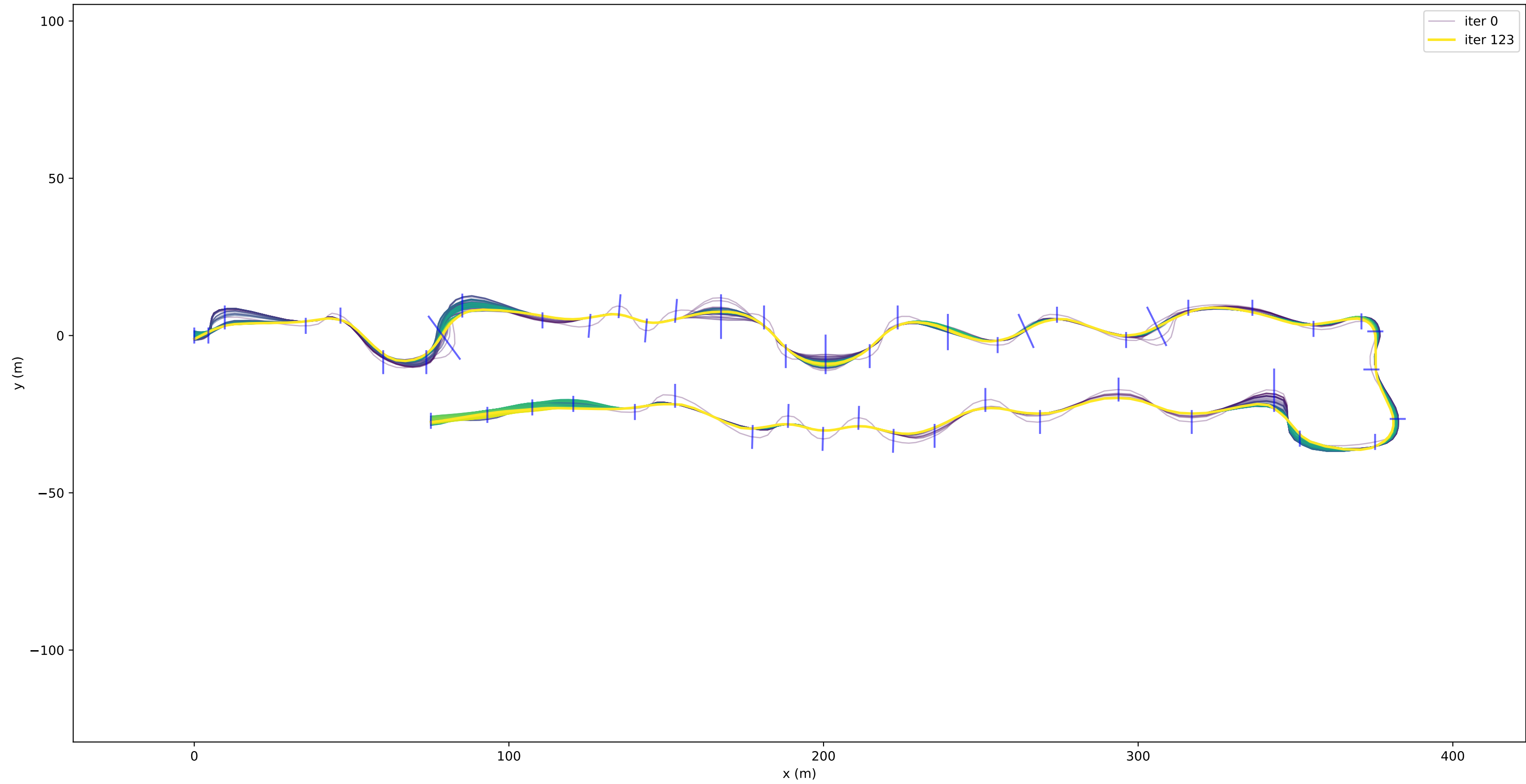
total objective function value: 1.786495

Per-step time estimate (one measured objective evaluation)

make spline from dofs: 6.77 ms
path length function: 0.51 ms
equidistant times generated: 0.33 ms
curvature objective function: 113.72 ms

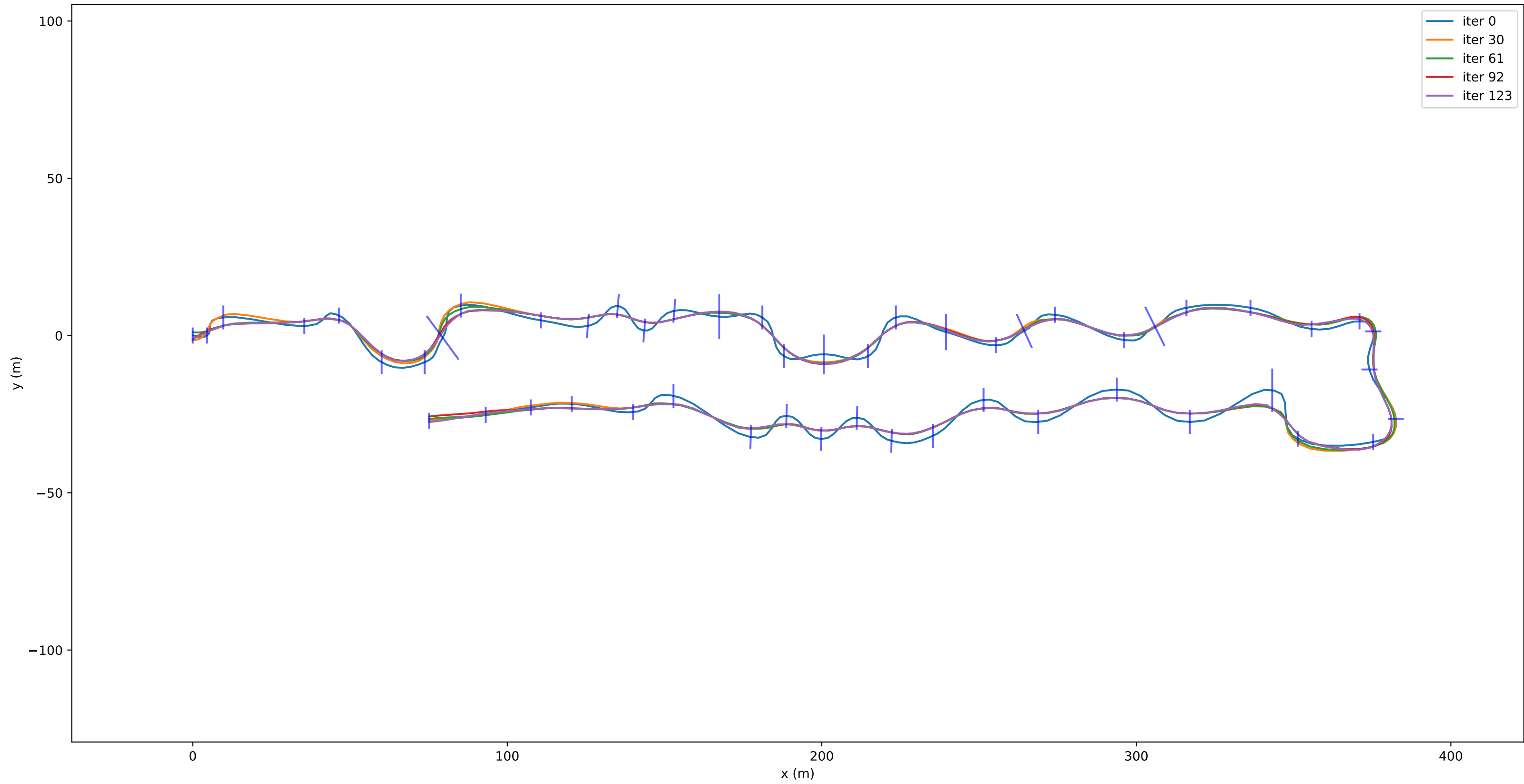
Optimization progression

Optimization progression (124 iterations)



Equally spaced iterations

5 equally spaced iterations



Instant radius vs distance

